

Definitions and Values

Name: _____

Date: _____

As a reminder, these are some constructors provided by the book:

Circle(int radius, String color) Rect(int width, int height, String color)

Q1. Does each line run? If not, say what is wrong in a few words.

definition	runs?	if not, what is wrong
Pict box = new Rect(20, 30, "blue");		
string name = "Val";		
Pict p = Circle(25, "red");		
int count = "5";		
Pict dot = new Circle(25);		
Pict p = new Circle("red", 25);		

Q2. For each of the two definitions below, label all the parts you can with **type**, **name**, **value**, **constructor**, **argument**

Pict dot = new Circle(25, "red");

int year = 2026;

Q3. Write a definition for a green rectangle 10 wide and 60 tall, named stem.

Q4. Write the value each name holds after this program runs (it doesn't have errors)

```
int total = 7 * 3;
int half = total / 2;
String label = "Total: " + total;
int agePlusOne = 17 + 1;
String textPlusOne = "17" + 1;
```

total = _____
half = _____
label = _____
agePlusOne = _____
textPlusOne = _____

Q5. Two students average the scores 7 and 8. Neither line is an error.

int mean1 = (7 + 8) / 2; double mean2 = (7 + 8) / 2.0;

mean1 = _____ mean2 = _____ Which belongs in a grade report? _____

Q6. In "Does each line run", Java would report each of the errors and refuse to run those programs. Would you have rather gotten an error from the mean example? Why or why not?